Can Selfish Agents Learn to Share?

Emergent Cooperation in Multi-Agent Pandemic Resource Allocation

Brooke Ballhaus Riya Narain

Department of Computer Science, Stanford University

Problem: Selfish Agents in a Shared Crisis

During a pandemic, scarce medical resources (ventilators, vaccines, PPE) must be allocated across many cities with **heterogeneous, asynchronous demand surges**. A central planner with full information could route resources optimally, but in practice cities act as **self-interested agents** controlling their own stockpiles.

> **Core Question:** Can RL agents trained with only local reward signals learn to *cooperate* by transferring resources to cities that need them more, rather than hoarding?

Why it matters: Resource hoarding during COVID-19, Ebola, and flu outbreaks caused preventable deaths. If RL can learn cooperative allocation without centralized control, it offers a scalable policy tool for future crises.

Prior Work & Gaps

Classical approaches (integer programming, proportional-to-need heuristics) assume a benevolent central planner and do not generalize to decentralized settings with strategic agents.

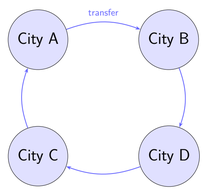
Multi-agent RL has achieved cooperation in symmetric games (Marris et al., 2021; Lowe et al., 2017), but pandemic resource allocation introduces two key complications:

- **Asynchronous crises:** cities hit peak demand at different times, so the value of a resource depends on when and where it arrives
- **Reward misalignment:** local death counts give no signal for sending resources away from a city that currently has surplus

Our contribution: A carefully shaped *impact reward*, combined with shared observations, is sufficient to teach decentralized agents genuine cooperation across 2–8 cities and multiple resource types.

Environment: Multi-City SEIR Pandemic

Simulation: $N \in \{2, 4, 6, 8\}$ cities, each running a calibrated SEIR model with staggered 30-day demand surges ($2\times$ transmission multiplier per city). Agents control weekly ventilator (and optionally vaccine / PPE) transfers via a token-market mechanism.



Observation (rich): infection rates, hospitalization loads, and resource stockpiles for *all* cities.
Scarcity: weekly supply of 5×10^{-5} ventilators per capita (tight) to 2.5×10^{-4} (moderate).
Episode length: 180 days; transit time 1 day.

References

1. Schulman et al. (2017). *Proximal Policy Optimization.* arXiv:1707.06347.
2. Yu et al. (2022). *The Surprising Effectiveness of PPO in Cooperative MARL.* NeurIPS.
3. Lowe et al. (2017). *Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments.* NeurIPS.
4. Marris et al. (2021). *Multi-agent training beyond zero-sum with correlated equilibrium meta-solvers.* ICML.
5. Mnih et al. (2015). *Human-level control through deep reinforcement learning.* Nature.
6. Hollingsworth et al. (2011). *Mitigation strategies for pandemic influenza.* J. Royal Soc. Interface.

Approach: Impact Reward + Decentralized RL

Key insight: Standard local rewards (own-city deaths) give no incentive to transfer. We replace them with an **impact reward** that credits a transfer only up to the recipient's current shortfall:

$$r_i^{\text{impact}} = \sum_{j \neq i} \min\left(\Delta_j,\ \text{transfer}_{i \to j}\right)$$

where Δ_j is city j's unmet ventilator demand. Sending more than a city needs earns nothing, which stops agents from dumping surplus and aligns individual incentives with global need.

We evaluate **IPPO**, **MAPPO**, **IPPO+peer-incentive**, **DQN**, and a **proportional-to-need heuristic** (oracle upper bound) across 8 ablations (E0–E7), progressing from local rewards to the full impact reward design with multi-resource and scarcity sweeps.

Key Results

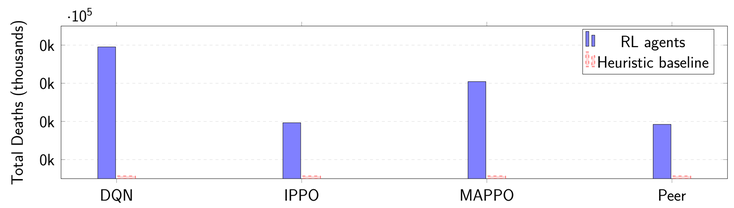
Default setting (4 cities, ventilators only):

Algorithm	Deaths (mean)	Δ vs. Heuristic	Gini	Welfare
Heuristic (oracle)	**162,765**	—	0.031	−163,578
Peer	216,905	+33.3%	0.074	−232,868
IPPO	218,589	+34.3%	0.049	−235,022
MAPPO	261,754	+60.8%	0.132	−290,268
DQN	298,096	+83.2%	0.090	−336,780

All values are means over 5 seeds. Heuristic has access to global need; RL agents are decentralized.

IPPO and Peer come closest to the oracle, within 33–34% on deaths. MAPPO surprisingly underperforms: its centralized critic is slower to train and the impact reward conflicts with its already team-oriented objective. DQN lags furthest because its coarse discrete action space cannot target transfers precisely.

RL vs. heuristic oracle (default setting):



Scaling (cities sweep, IPPO vs. DQN):

	Deaths (mean)		Gini	
Cities	IPPO	DQN	IPPO	DQN
2	134,928	175,056	0.033	0.056
4	218,589	298,096	0.049	0.090
6	281,446	354,191	0.056	0.106
8	337,982	410,640	0.077	0.077

IPPO outperforms DQN at every scale and stays more equitable throughout. Both methods get harder to coordinate as cities increase, but IPPO degrades more gracefully.

Additional Findings

E5: Multi-resource (ventilators + vaccines + PPE)

Adding vaccines and PPE yields **6–11% further death reductions**. Peer incentives show the best equity here: the token market prices each resource by urgency, so cities that need help most get it first, without any retuning.

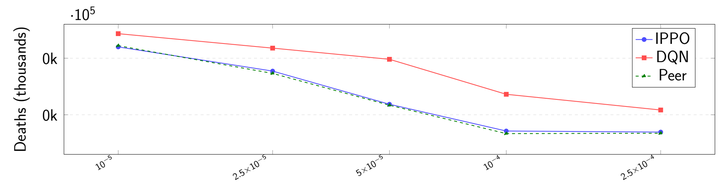
E6: Exchange-rate robustness

Token exchange rates swept over a $25\times$ range. Deaths varied only 198k–207k with overlapping 95% CIs. The peer-incentive mechanism is **robust** and does not need careful hyperparameter tuning to work.

E7: Transfer cost scope condition

A 30% transfer cost causes agents to revert to hoarding. Cooperation only emerges when transfers are cheap, which sets a clear boundary for where this approach applies in practice.

Scarcity sweep (weekly supply per capita):



IPPO and Peer beat DQN at every supply level. Peer pulls ahead of IPPO under tight supply, where the token market helps most. The gap shrinks under abundance, when even selfish agents do reasonably well.

Conclusions, Limitations & Future Work

> **(1)** Impact reward + rich observations are **sufficient to teach decentralized agents genuine cooperation** across 2–8 cities and multiple resource types.
> **(2)** Policy-gradient methods (IPPO, Peer) consistently outperform DQN; the cooperation mechanism is **policy-gradient-specific**.
> **(3)** Cooperation is **fragile to transfer costs**: a 30% cost causes agents to revert to hoarding entirely.
> **(4)** Peer incentives achieve the best equity in multi-resource settings, suggesting token markets as a practical coordination tool.

Limitations:

- The heuristic still outperforms all RL methods, likely because it has access to global information that agents do not
- The SEIR model omits hospital capacity limits and within-city variation
- Agents are identical; real cities have different political power and supply contracts

Future Work:

- Understand when during training cooperation emerges
- Replicate on real CDC city-level data
- Test with asymmetric agents and political bargaining